

Abhishek Agrawal

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PROFESSIONAL SUMMARY

Data Engineer (5 yrs) who builds metadata-driven, declarative data platforms that turn a SQL spec into governed, production ETL. At NBCUniversal, architected a data fabric that auto-generates pipelines, infrastructure, workflow orchestration across AWS, Databricks, & Snowflake – with built-in cataloging, lineage, GDPR/CCPA governance, zero-copy sharing & cost/SLA-aware compute. Delivered a terabyte-scale Olympics 2024 ad-impression reporting platform, beating a hard 2-hour SLA at under 1 hour. Extends into generative modeling, RAG, lakehouses, and vector Databases.

TECHNICAL SKILLS

Programming: Python, SQL, Scala, Java, PySpark, Shell/Bash, JavaScript

Ingestion | Orchestration: Apache Spark, Kafka, Flink, Airflow, dbt, ETL/ELT, Batch & Streaming, Event-Driven Pipelines

Cloud | Governance: AWS, GCP, Snowflake, Databricks, Redshift, Glue/Unity Catalog, Delta Lake, Apache Iceberg, Parquet, Great Expectations, RBAC/IAM

Architectures | Modeling: Data Fabric, Lakehouse, Warehousing, Metadata & Graph Catalog, Zero-Copy Sharing, MPP, Dimensional/Normalized Modeling

DataOps | Platform Engineering: Terraform, CloudFormation, Docker, Kubernetes, CI/CD, Git, CloudWatch, Flask/FastAPI REST APIs, React

AI/ML | Retrieval: PyTorch, RAG, Vector Search & Embeddings, MLOps, Feature Engineering, Text-to-SQL, LLM Integration, Scikit-learn, Hugging Face

EXPERIENCE

Research Assistant – New York University, US

Dec 2025 – May 2026

HPC Telemetry Lakehouse & Power-Forecasting Pipeline · NYU HPC, GCP, Python, PyTorch, PySpark, Parquet

- Architected a medallion lakehouse over ~2 TB of MIT SuperCloud telemetry – 460K+ compute jobs (98K GPU), V100 GPUs, 9 months – resampled, aggregated, feature-engineered with AQE, partitioning & skew handling on 16-node cluster to cut end-to-end latency ~80%.
- Engineered a 2-stage diffusion system – SLURM job-metadata synthesis feeding a sliding-window autoregressive transformer (DDIM-50+CFG) – generating conditioned GPU power traces over 4,400+ test jobs (peak Corr = 0.93) for datacenter power forecasting and capacity planning.

Data Engineer (Contractor) - NBCUniversal, India

Aug 2021 – Aug 2024

- Automated QA, governance & deployment (CI/CD, Great Expectations, pytest) - adopted by other teams as their standard framework.
- Partnered with stakeholders on architecture & SLAs, delivering modular-monolith APIs & React UIs with microservice-based ETL integrations.
- SME across AWS, Snowflake, Airflow & CI/CD – drove bootcamps; amid team restructuring, absorbed core modules & new out-of-scope work.

Enterprise Data Fabric · AWS (S3, Lambda, EMR/EMR Serverless, Glue, Redshift, Step Functions, EventBridge, SQS/SNS, ECS/Fargate, RDS, CloudWatch), Snowflake, Databricks, Airflow, Terraform, Python, Scala/PySpark, Kafka, Flink, Flask, React, PostgreSQL, Iceberg, Delta, Parquet

- Architected a declarative data platform that turns a SQL spec into production data products – auto-provisioning ingestion, transformation, orchestration, & infrastructure(IaC) from metadata – collapsing new-pipeline delivery from weeks to hours (~75% faster time-to-market).
- Engineered a graph-modeled metadata catalog that maps cross-domain ontologies across siloed sources – delivering a unified business glossary, automated lineage, and policy-enforced RBAC with GDPR/CCPA controls – enabling governed, self-service data discovery across the organization.
- Designed an SLA-driven compute optimizer that dynamically distributes workloads across Lambda, EMR, EMR Serverless, Databricks, and Snowflake – provisioning a target SLA or defaulting to the lowest-cost plan – reducing infrastructure spend ~20%.
- Automated per-request generation of Airflow DAGs & Step Functions – compiling dependencies, scheduling, & full-refresh vs. incremental state into ready-to-run workflows – processing ~75 TB of ad-impression data with end-to-end observability.
- Operationalized zero-copy data sharing across accounts (Snowflake Shares, S3 cross-account) – delivering live, single-source data to external partners and internal ML teams with no duplication, governed by IAM, ACLs, and grants that prevent copy & exfiltration.

Olympics 2024 Serverless ETL Migration · AWS (EMR Serverless, Lambda, S3, Redshift), Airflow, Python, Scala/Spark, Cassandra, Databricks

- Migrated Olympics 2024 ad-impression reporting from on-prem Cassandra & on-demand Databricks compute to a fully serverless AWS stack under a hard delivery deadline, governed by CI/CD.
- Re-architected fragmented, sequential, I/O-bound legacy jobs into a single config-driven Airflow framework with dynamically parallelized Spark and Lambda tasks – sustaining a twice-daily full refresh of 60+ datasets across linear and digital sources.
- Reduced end-to-end refresh from ~2 hours to under 1 hour (~50%), clearing the 2-hour SLA with headroom and full observability (CloudWatch, SNS).

Schema-Grounded Enterprise Retrieval (RAG) · Azure OpenAI, Hugging Face (KeyBERT), Python, React, Flask, PostgreSQL

- Built a schema-grounded RAG assistant (React/Flask, CI/CD) that unifies text-to-SQL and document Q&A by retrieving over a graph metadata catalog (KeyBERT-indexed) – grounding SQL to real tables/columns and citation-backed (docs/wikis/PDFs) answers.
- Hardened accuracy with SQL validation before execution and event-driven metadata refresh to counter knowledge drift – served through one RBAC-governed interface, cutting analyst self-service from ~30 minutes to under a minute.

Additional Experience – Teaching Assistant, NYU (US) · SDE Freelance & Internships, India · 2020 – 2025

PROJECTS

Credit Risk GPT · AWS S3, Databricks, Lakeflow Declarative Pipelines, Unity Catalog, PySpark, Claude, OpenAI

- Built a credit-risk analytics platform over ~2.26M Lending Club loans – a governed Databricks lakehouse (Lakeflow Declarative Pipelines, Unity Catalog lineage/RBAC) materializing risk KPIs by grade, vintage, geography, and DTI.
- Extended it with a retrieval-augmented (RAG) analyst assistant that grounds credit-risk Q&A in CFPB regulatory complaint narratives alongside the governed metrics – surfacing consumer-protection and compliance signals that raw loan data alone misses.

NYC Housing Pulse(2015–2025) · AWS S3, QuickSight, Python

- Engineered an S3-backed batch pipeline ingesting ~900K NYC Dept. of Finance property-sale transactions across all 5 boroughs (2015–2025), resolving inconsistent schemas, malformed addresses, and duplicate parcels into a clean, conformed serving layer.
- Transformed raw sales into borough- and year-level analytics — price-per-sqft, sales volume, and affordability trends — powering a QuickSight dashboard that traced a decade of market shifts across neighborhoods and price tiers.

EDUCATION

New York University, US

Aug 2024 – May 2026

MS, Computer Engineering; GPA: 3.5/4

Nagpur University(RCOEM), IN

Aug 2017 – May 2021

BS, Computer Engineering; GPA: 3.5/4